

Simple Exercise: Learn React States for Beginners

Objective

Introduce beginners to **React state** by creating a simple "Toggle Message" app where a message is shown or hidden using a button.

Explanation of React State

In React, **state** is used to store information about a component that can change over time.

- **useState Hook:** A special React function used to create and manage state in functional components.

Syntax:

```
const [state, setState] = useState(initialValue);
```

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- **state:** The current value of the state.
 - **setState:** A function to update the state.
 - **initialValue:** The starting value of the state.
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Exercise: Create a Toggle Message App

Steps

1. Create a new component: **ToggleMessage.js**.
 2. Use **useState** to manage whether the message is shown or hidden.
 3. Add a button to toggle the message on or off.
 4. Display the component in **App.js**.
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Solution

Step 1: Create **ToggleMessage.js**

```
import React, { useState } from 'react';
```

```
function ToggleMessage() {
  // Step 2: Define state for toggling the message
  const [isVisible, setIsVisible] = useState(false);

  // Step 3: Function to toggle the state
  const toggleMessage = () => {
    setIsVisible(!isVisible); // Flips the state between true and false
  };

  // Step 4: Render the UI
  return (
    <div>
      <h1>Toggle Message App</h1>
      <button onClick={toggleMessage}>
        {isVisible ? 'Hide Message' : 'Show Message'}
      </button>
      {isVisible && <p>Hello, React State is Awesome!</p>}
    </div>
  );
}

export default ToggleMessage;
```

Step 2: Update **App.js**

```
import React from 'react';
import ToggleMessage from './ToggleMessage';

function App() {
  return (
    <div>
      <ToggleMessage />
    </div>
  );
}

export default App;
```

What Happens in the Code?

1. State Initialization:

- `const [isVisible, setIsVisible] = useState(false);`
 - `isVisible` is the state variable (initially `false`).
 - `setIsVisible` updates `isVisible`.

2. State Toggle:

- The `toggleMessage` function flips the value of `isVisible` when the button is clicked.

3. Conditional Rendering:

- `{isVisible && <p>Hello, React State is Awesome!</p>}`:
 - The `<p>` tag is shown only when `isVisible` is `true`.

Expected Output

Initially displays:

Toggle Message App

[Show Message]

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Clicking **Show Message** displays the message:

Toggle Message App

[Hide Message]

Hello, React State is Awesome!

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- Clicking **Hide Message** hides the message again.

What Students Will Learn

1. `useState` Basics:

- Declaring and updating state.

2. Event Handling:

- Using `onClick` to trigger state updates.

3. Conditional Rendering:

- Dynamically showing/hiding elements using state.

This is a beginner-friendly exercise perfect for understanding state with minimal complexity!

